Technical Datasheet — V4b "h60 + anode margin" (VBF-T1R2-DSH-01)

Next-generation pure electric sedan cell — NMC811 / graphite (Chen2020 parameter set).

Field	Value	Source
Rated capacity	5.0 Ah (nominal); 5.6997 Ah (simulation-verified 1C discharge)	parameter set `Nominal cell capacity [A.h]`; `cell/r4_v4b_1c_spme.json:capacity_ah`
Nominal voltage	4.086 V (1C midpoint)	`cell/r4_v4b_energy.json:midpoint_voltage_v`
Voltage window	2.5 – 4.2 V	parameter set `Lower/Upper voltage cut-off [V]`
Rated energy	20.55 Wh	`cell/r4_v4b_energy.json:energy_wh`
Energy density	583.2 Wh/kg (contract caliber, electrolyte excluded); 478.7 Wh/kg incl. electrolyte estimate	`cell/r4_v4b_energy.json:energy_density_wh_kg`; electrolyte-incl. `inferred` (20.553 Wh / 0.04293 kg, electrolyte density 1.2 g/cm ^{3} literature `estimate`)
Volumetric energy density	1028.4 Wh/L	`cell/r4_v4b_energy.json:energy_density_wh_l`
DCR	80.4 μΩ	`cell/r4_v4b_energy.json:dcr_ohm` (1C-point estimate)
Power density	1498.5 kW/kg	`cell/r4_v4b_energy.json:power_density_w_kg` (contract-caliber formula)
Maximum continuous discharge rate	1C (5 A) — simulated	`cell/r4_v4b_1c_spme.json`
Fast-charge capability	4C CC charge: T_max 326.5 K, no lithium plating (anode potential min +0.0215 V), CC acceptance 4.53 Ah (79% SOC)	`cell/r4_v4b_4c_dfn.json` (`T_max_K`, `anode_potential_v` min); acceptance `inferred`: 815 s × 20 A / 3600
Operating temperature range	Simulated at 298.15 K ambient (25 °C); 4C fast charge verified to cell T_max 326.5 K with h = 60 W/m ^{2} · K cooling. Low-temperature performance: not simulated in this case	protocol `T_amb_K`; sim outputs; honest scope statement
Cycle life	Not simulated (durability not in this case's criteria; the Chen2020 set is aging-capable — SEI kinetic rate constant 1e-12 m/s — and the aging protocol is available for a follow-up round). Must not be read as a cycle-life claim	—
Safety	4C: plated = false, T_max ≤ 333.15 K pass. Overcharge to 4.7 V (0.5C): T_max 299.4 K, thermal-runaway trigger = false (lumped 0-D model). Nail/crush/drop: N/A (beyond pure simulation boundary, requires physical experiment)	`cell/r4_v4b_4c_dfn.json`, `cell/r4_v4b_overcharge.json`, `cell/r4_v4b_tr.json`
Dimensions	Electrode area 0.1027 m ^{2} ; electrode stack thickness 194.6 μm; design cell volume 2.514e-5 m ^{3} (thermal sim). Height × width: Not provided (cell can not modeled)	`cell/r4_v4b_energy.json`; `cell/params_v4b.json`
Mass	35.24 g (contract caliber, electrolyte excluded); 42.93 g incl. electrolyte estimate	`cell/r4_v4b_energy.json:mass_kg`; § 3/ § 4 of design_spec.md

Evidence: all safety/performance determinations above are mechanical `bda log-evaluate` round-4 verdicts (pass), recorded in `log.jsonl`.